Senior High School

DEPARTMENT OF EDUCATION

Learning Activity Sheet for General Mathematics

Quarter 4

Unit

12

Learning Activity Sheet for General Mathematics
Quarter 4: Unit 12

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LEARNING ACTIVITY SHEET

Learning Area:	General Mathematics	**Quarter:**	4
Lesson No.:	12.1	**Date:**	
Lesson Title/ Topic:	Logical Propositions		
Name:		**Grade & Section:**	

I. **Activity: Building Blocks of Logic**

II. **Objective(s):**

1. Recognize whether a statement is a proposition or not.
2. Distinguish between simple and compound propositions.
3. Identify the logical operator(s) present in a given proposition.
4. Apply various logical operations to propositions including constructing truth tables.

III. **Materials Needed:** Activity sheets (with Parts A–D: propositions, simple/compound, operators, truth tables), and pen/pencil

IV. **Instructions:**

In this activity, you will practice working with propositions and logical operators. First, you will identify whether a statement is a proposition and decide if it is true or false. Then, you will classify statements as either simple or compound and spot the logical operators used. Finally, you will apply what you learned by writing propositions in symbols and creating truth tables. This activity will help you master how to analyze and combine statements in logic step by step.

A. Spot the Proposition

Directions: For each statement, decide whether it is a proposition or not. If it is a proposition, state its truth value (True or False).

1. The Earth revolves around the Sun.
2. What time is your class tomorrow?
3. Close the window.
4. 2 + 5 = 8.
5. Manila is the capital of the Philippines.
6. Do your homework.

Guide Questions:

1. Is it a declarative sentence?
2. Can it be clearly labeled as **true** or **false**?

B. Sorting Simple and Compound

Directions: Classify the following as **Simple** or **Compound Propositions**. If compound, identify the type of connective(s) used.

1. The sky is clear.
2. The sky is clear and the air is fresh.
3. If it rains, then the ground will be wet.
4. Either the lights are off or the power is interrupted.
5. Jose Rizal is a national hero.
6. The school is open if and only if it is Monday.

C. Operator Hunt

Instructions: Identify the **logical operator(s)** used in each proposition below and rewrite them using symbols.

Expected Output Example:

- Sentence: "It is not true that the number is even."
- Operator: Negation (~)
- Symbolic Form: ~p

1. It is not true that the number is even.
2. The students are noisy and the teacher is absent.
3. If the bell rings, then the class will be dismissed.
4. You will pass if and only if you study.
5. Either the exam is today or it is tomorrow.

D. Truth Table Challenge

Directions: Construct the truth tables for the following compound propositions:

1. $\sim p \vee q$
2. $(p \wedge q) \rightarrow \sim p$
3. $(p \vee q) \leftrightarrow (\sim p \wedge q)$

Guide Questions:

a. How many rows will the table have for 2 variables?
b. What is the truth value of the compound statement in each case?

V. Synthesis:

1. How did this activity help you understand what makes a statement logical or not?

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2. What clues helped you decide whether a statement is true or false?

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3. How did you determine the relationships or connections between different propositions?

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4. Why are logical operators important in forming clear and meaningful statements?

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5. How can using logic help you make better decisions or avoid false information in real life?

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I. Activity: Assessment on Logical Proposition

II. Objective(s):

1. Recognize whether a statement is a proposition or not.
2. Distinguish between simple and compound propositions.
3. Identify the logical operator(s) present in a given proposition.
4. Apply various logical operations to propositions including constructing truth tables.

III. Materials Needed: Activity Worksheet, Pen/Pencil

IV. Instructions:

In this activity, you will show what you have learned about logical propositions and operators. You will evaluate given statements, decide if they are propositions or not, classify them as simple or compound, and translate them into symbols. You will also practice combining two or more statements using logical connectors such as and (∧), or (∨), not (~), if…then (→), and if and only if (↔). Finally, you will construct truth tables to test whether compound statements are true or false. This will help you check how well you understand logic and how to apply it to reasoning clearly.

A. True or False. Determine if each statement is True or False.

1. Baguio is the capital of the Philippines.
2. 10 is an even number.
3. All squares are rectangles.
4. The moon is made of cheese.
5. If it rains, then the ground will be wet.

B. Proposition or Not. State whether each is a Proposition or Not a Proposition. If it is a proposition, determine its truth value (True or False).

6. What is your favorite subject?
7. The Earth has one moon.
8. Please close the window.
9. 5 + 7 = 13.
10. The Pacific Ocean is larger than the Atlantic Ocean.

C. Simple or Compound. Classify each as a Simple Proposition or Compound Proposition.

11. A triangle has three sides.
12. It is raining and the wind is strong.
13. If you study well, then you will pass the exam.
14. The classroom is clean.
15. Jose Rizal is the national hero or Andres Bonifacio is the national hero.

D. Using Logical Operators

Express the following in **symbols**. Let:

p: The sky is blue.

q: It is daytime.

r: The sun is shining.

16. If it is daytime, then the sun is shining.
17. The sky is not blue.

18. The sun is shining or it is daytime.
19. The sky is blue if and only if it is daytime.
20. $(p \leftrightarrow q) \wedge {\sim}r$

E. Truth Table. Construct the truth table for the compound proposition.

$$(p \wedge {\sim}q) \rightarrow ({\sim}p \leftrightarrow q)$$

V. Synthesis:

1. What steps did you follow to decide whether a statement is true or false and whether it is a proposition?

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2. How can you tell a declarative sentence (proposition) apart from a question or a command? Give one example from the worksheet.

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3. How did combining simple statements with words like and, or, if…then, or if and only if change the meaning of the original statements?

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4. How did writing sentences with symbols (p, q, r, $\sim$, $\wedge$, $\vee$, $\rightarrow$, $\leftrightarrow$) help you understand their logical relationships?

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5. What did the truth table show you about the overall truth of the compound statement — when is a compound always true, always false, or sometimes true?

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LEARNING ACTIVITY SHEET

Learning Area:	General Mathematics	**Quarter:**	4
Lesson No.:	12.2	**Date:**	
Lesson Title/ Topic:	Conditional Propositions		
Name:		**Grade & Section:**	

I. **Activity: Transform it: Constructing the converse, inverse, and contrapositive of a conditional statement**

II. **Objective(s):**

1. Construct the converse, inverse, and contrapositive of a given conditional statement using sentences and symbols.

III. **Materials Needed:** Writing Paper, Pen

IV. **Instructions:**

Construct the converse, inverse, and contrapositive of the following conditional propositions in words and symbols.

1. Conditional Statement: If the internet connection is stable, then online classes run smoothly.

   Converse: ____

   ____

   Inverse: ____

   ____

   Contrapositive: ____

   ____

2. If someone throws garbage in the river, then the water becomes polluted.

   Converse: ____

   ____

   Inverse: ____

   ____

   Contrapositive: ____

   ____

V. **Synthesis:**

1. How did you transform a conditional statement to its converse, inverse, and contrapositive?

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2. What part of the activity do you find difficult? Why?

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3. How would you apply conditional statements and its forms in real-life situations?

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LEARNING ACTIVITY SHEET

Learning Area:	General Mathematics	**Quarter:**	4
Lesson No.:	12.3	**Date:**	
Lesson Title/ Topic:	**Syllogisms: Modus Ponens and Modus Tollens, Inverse and Converse Fallacy**		
Name:		**Grade & Section:**	

I. **Activity: "Logic Detectives"** 🔍

II. **Objective(s):**

1. **Identify** whether a given reasoning pattern follows *Modus Ponens*, *Modus Tollens*, *Fallacy of the Converse*, or *Fallacy of the Inverse* with accuracy.
2. **Explain** the reasoning behind each case by providing a brief justification that shows understanding of valid and invalid argument structures
3. **Construct** their own examples of reasoning statements that demonstrate both valid and fallacious logic patterns, applying them to real-life or contextual situations.

III. **Materials Needed:** Printed quiz sheet/ projector display of the activity, answer sheets or notebooks, pen

IV. **Instructions:**

1. Read each case carefully.
2. Identify the reasoning pattern used.
3. Write the correct label (Modus Ponens, Modus Tollens, Fallacy of the Converse, Fallacy of the Inverse).
4. Justify your answer in one sentence.

Scenario for Students:

You are now *Logic Detectives*. Each case contains a reasoning statement. Your mission is to analyze the case and decide whether the reasoning follows ***Modus Ponens, Modus Tollens, Fallacy of the Converse, or Fallacy of the Inverse.***

☑Valid- Modus Ponens and Modus Tollens

✕ Invalid- Fallacy of the Inverse and Fallacy of the Converse

CASES!

1. **Case of the Rainy Day** ☂
   If it rains, then the ground will be wet.
   It rains.
   Therefore, the ground is wet.
   👉 Reasoning Type: __________

2. **Case of the Failed Exam**
   If a student studies hard, then she will pass the test.
   She did not pass the test.
   Therefore, she did not study hard.
   👉 Reasoning Type: __________

3. **Case of the Traveler** ✈
   If I am in Cebu, then I am in the Philippines.
   I am in the Philippines.
   Therefore, I am in Cebu.
   👉 Reasoning Type: __________

4. **Case of the Musician** 🎹
   If I practice piano daily, then I will become a good pianist.
   I did not practice piano daily.
   Therefore, I will not become a good pianist.
   👉 Reasoning Type: __________

5. **Case of the Athlete** 🏀
   If I exercise regularly, then I will be physically fit.
   I am physically fit.
   Therefore, I exercise regularly.
   👉 Reasoning Type: __________

6. **Case of the Pet Lover** 🐶
   If someone owns a dog, then they love animals.
   This person does not own a dog.
   Therefore, they do not love animals.
   👉 Reasoning Type: __________

7. **Case of the Light Bulb** 💡
   If the switch is on, then the light will shine.
   The light is not shining.
   Therefore, the switch is not on.
   👉 Reasoning Type: __________

Guide Questions:

1. Among all the cases, which reasoning pattern do you think is the most common in everyday situations? Why do you think so?
   ____
   ____
   ____

2. How can identifying valid and invalid reasoning help you make better decisions in real life?
   ____
   ____
   ____

3. Choose one case and change it so that it becomes a valid reasoning pattern. Explain your revision.

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4. In social media or news posts, do you think people often use fallacious reasoning? Give an example and explain what kind of fallacy it might be.

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5. Reflect on your experiences in school or at home. When have you or someone else used reasoning similar to Modus Tollens? What was the situation and the conclusion?

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6. Why do you think it's important for students to learn about logical reasoning patterns like Modus Ponens and Modus Tollens?

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Rubric for Assessment:

Criteria	**4 – Excellent**	**3 - Proficient**	**2-Developing**	**1-Beginning**
Accuracy (Correct identification of reasoning type)	Identifies all cases correctly	Identifies 5–6 correctly	Identifies 3–4 correctly	Identifies 0–2 correctly
Explanation/ Justification	Gives clear, logical explanations for all answers	Gives clear explanations for most answers	Gives partial or unclear explanations	Gives little or no explanation
Participation/ Engagement	Actively participates, collaborates well, and shows enthusiasm	Participates with some prompting	Limited participation, needs encouragement	Passive, little to no participation

Scoring Guide:
10–12 points = Excellent Mastery
7–9 points = Proficient

4–6 points = Developing
1–3 points = Beginning

I. **Activity: Reason it Right!**

II. **Objective(s):**

1. Identify and classify different types of reasoning- Modus Ponens, Modus Tollens, Fallacy of the Inverse, and Fallacy of the Converse.
2. Construct own examples demonstrating both valid and fallacious forms of reasoning.
3. Explain the logical process they used in analyzing and answering reasoning problems.
4. Reflect on the importance of applying logical reasoning in interpreting real-life statements and situations

III. **Materials Needed:** Printed quiz sheet/ projector display of the activity, answer sheets or notebooks, pen

IV. **Instructions:**

Read and follow each part carefully.
For Part A, identify the type of reasoning shown in each statement—Modus Ponens, Modus Tollens, Fallacy of the Inverse, or Fallacy of the Converse.
For Part B, create your own examples for each reasoning type, making sure they are clear and realistic.
In Part C, answer the reflection questions thoughtfully, explaining how you applied reasoning and what you learned. Be original, write neatly, and submit your work on time.

Part A: Identification

Tell whether the reasoning is **Modus Ponens, Modus Tollens, Fallacy of the Inverse, or Fallacy of the Converse.** (1 point each)

1. If it is Saturday, then we don't have class. It is Saturday. Therefore, we don't have class.

2. If a student studies hard, then they will pass the test. The student did not pass the test. Therefore, they did not study hard.

3. If an animal is a dog, then it is a mammal. This animal is a mammal. Therefore, it is a dog.

4. If a person is a teacher, then they can speak in class. This person is not a teacher. Therefore, they cannot talk in class.

5. If the lights are on, then someone is home. The lights are not on. Therefore, no one is home.

Part B: Application

Write **your own example** (1 each) for:

1. Modus Ponens (valid)

2. Modus Tollens (valid)

3. Fallacy of the Inverse (invalid)

4. Fallacy of the Converse (invalid)

Part C. Synthesis:

a. How did you arrive at your answers?

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b. How did you find the activity? Can you share and describe your experience?

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c. How did understanding different forms of reasoning (Modus Ponens, Modus Tollens, Fallacy of the Inverse, and Fallacy of the Converse) make it easier to analyze real-life statements or situations? Can you think of a real-life scenario where applying correct reasoning would be helpful?

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d. What's the most important thing you learned from this activity about identifying valid and fallacious reasoning? How can this help you make better judgments in real-life situations?

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